Differences in Input and Output Quality of LLMs Across Age Groups (#246249)

Author(s)
This pre-registration is currently anonymous to enable blind peer-review.
It has 2 authors.

Pre-registered on:
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1) Have any data been collected for this study already?
No, no data have been collected for this study yet.

2) What's the main question being asked or hypothesis being tested in this study?
The study will investigate whether individuals from different age groups differ in how they formulate prompts to large language models (LLMs), and whether such differences affect the quality and effectiveness of the model's responses.
Input differences:
Users from different age groups will differ in how they formulate prompts. Specifically, prompt length, linguistic complexity, politeness, and structural clarity are expected to vary between younger and older users.
Output differences:
The quality and effectiveness of the model's responses will vary depending on the age group of the user who generated the prompt, with prompts from younger users expected to elicit higher-quality responses.

3) Describe the key dependent variable(s) specifying how they will be measured.
Participants will rate each model response on four aspects (addressing the question, usefulness, clarity, and sufficiency) using a 5-point Likert scale (1 = Not at all, 5 = Completely). Analyses may examine items separately or as a composite score, depending on reliability.
Additional dependent variables will include:
Response time (measured in hundredths of a second).
Error coding: number of factual errors and irrelevance errors in the model's outputs.
External quality evaluations: ratings by other LLMs (e.g., Gemini, Claude) and by independent human raters.

4) How many and which conditions will participants be assigned to?
Two between-subject conditions: younger adults (18–40) vs. older adults (65+).
Each participant will complete two within-subject tasks: a trip-planning task and a greeting-writing task.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.
We will first examine input differences by comparing prompt characteristics (e.g., length, linguistic complexity, politeness, structural clarity) between age groups using t-tests and regression analyses.
We will then use linear mixed-effects models to predict response quality, with Age group (younger vs. older) as a between-subjects factor and Task type (trip planning vs. greeting writing) as a within-subjects factor. Dependent variables will include self-ratings of response quality, model-based scores, and human judgments.
Finally, we will conduct exploratory regressions to test whether prompt characteristics predict response quality.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.
We will exclude participants who have incomplete data, experience technical failures, or produce model outputs with systematic shortcomings (e.g., empty, nonsensical, or technically invalid responses).
We will exclude participants whose total completion time is more than 55 minutes (i.e., those completing the study unusually slowly compared to the rest of the sample).
We will exclude participants whose performance does not meet the task requirements.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.
We will recruit a total of 84 participants (42 younger adults and 42 older adults). Participants will be recruited via Prolific using the following eligibility criteria: they must be native English speakers residing in the UK, with a minimum of 50 previously completed tasks and an approval rate of at least 98%. All participants will have prior experience using the internet, but no previous experience with AI systems such as GPT. Additional filters available on Prolific will be reviewed and applied as appropriate. This number is based on an a priori power analysis conducted with G*Power (α = 0.05, power = 0.95, f = 0.20).

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)
We will conduct exploratory analyses of linguistic and stylistic features of the prompts and responses (e.g., lexical indicators, structural patterns) to examine potential group differences and their relation to output quality. We will also use computational models for exploratory purposes to assess whether age groups can be distinguished based on these features. Additionally, we will also explore training a model to predict the writer's age group based on the prompts or outputs.